

Lecture 4:

Classification Analysis

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GRA4160: Predictive Modelling with Machine Learning

January 28, 2026

- ▶ **Recap** from last week: LDA and Regularized Regression
- ▶ **Logistic Regression**
- ▶ **Decision Trees**
- ▶ **Exercise: Recognizing handwritten digits**

Recap: Linear Discriminant Analysis (LDA)

► Purpose:

- Supervised classification & dimensionality reduction.
- Projects data onto a lower-dimensional space to maximize class separation.

► When to Use:

- Two or more well-separated classes with similar covariance matrices.
- Need for interpretable linear boundaries and reduced dimensionality.

► Mathematical Core:

- Defines scatter matrices: $\mathbf{S}_B$ (between-class) and $\mathbf{S}_W$ (within-class).
- Objective:

$$\max_{\mathbf{w}} \frac{\mathbf{w}^T \mathbf{S}_B \mathbf{w}}{\mathbf{w}^T \mathbf{S}_W \mathbf{w}}.$$

Recap: LDA Assumptions & Practical Tips

▶ Key Assumptions:

1. Each class follows (approximately) a normal (Gaussian) distribution.
2. All classes share the same covariance matrix.
3. Features are ideally uncorrelated, but mild correlations are often acceptable.

▶ Implications:

- ▶ If covariances differ significantly, consider Quadratic Discriminant Analysis (QDA).
- ▶ LDA can be robust to mild violations of normality.

▶ Practical Tips:

- ▶ Scale/standardize data to stabilize covariance estimation.
- ▶ Use cross-validation to evaluate classification performance (e.g., confusion matrices).
- ▶ Works well with moderate-to-large datasets that have reasonable class separation.

- ▶ **Takeaway:** A simple, interpretable linear boundary with dimensionality reduction under Gaussian assumptions.

Recap: Regularized Regression (Ridge, Lasso, Elastic Net)

- ▶ **Motivation:** Mitigate overfitting by penalizing large coefficients.
- ▶ **Methods Overview:**
 - ▶ **Ridge (L2):**
 - ▶ Shrinks coefficients (rarely to exactly zero).
 - ▶ Useful for handling multicollinearity.
 - ▶ **Lasso (L1):**
 - ▶ Can set some coefficients to zero (*feature selection*).
 - ▶ Often helpful in high-dimensional settings.
 - ▶ **Elastic Net (L1 + L2):**
 - ▶ Balances the strengths of Ridge and Lasso.
 - ▶ Two hyperparameters: overall strength (λ) + mixing ratio (α).
- ▶ **Choosing λ :**
 - ▶ Use cross-validation to find the best bias-variance trade-off.
 - ▶ Standardize features before applying the penalty terms.
- ▶ **Key Takeaway:** Regularization improves generalization, reduces overfitting, and can do feature selection (Lasso/Elastic Net).

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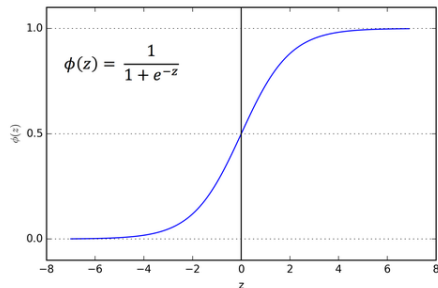
- ▶ The log-odds transformation (logit) ensures predicted probabilities are always in $[0, 1]$.
- ▶ Often fit by *maximum likelihood* (minimizing negative log-likelihood).
- ▶ Can be extended to multi-class problems (via One-vs-All or Multinomial/Softmax).

The Logistic (Sigmoid) Function

- Maps $\mathbb{R}$ to $[0, 1]$, producing valid probabilities.

$$f(z) = \frac{1}{1 + e^{-z}}$$

- As $z \rightarrow \infty$, $f(z) \rightarrow 1$; as $z \rightarrow -\infty$, $f(z) \rightarrow 0$.



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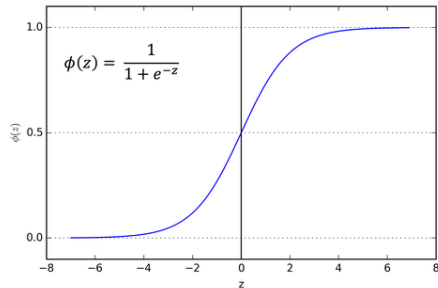
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Relates directly to the log-odds:

$$p = \frac{1}{1 + e^{-z}} = \frac{e^z}{1 + e^z}$$



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Relation: Log-odds, Probability, and Sigmoid

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$$p = \frac{1}{1 + e^{-z}} \quad (\text{the sigmoid})$$

- Changing z by 1 unit multiplies the odds $\frac{p}{1-p}$ by $e^1 \approx 2.718$.

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- ▶ A **non-parametric** method for classification and regression.
- ▶ The data is repeatedly *split* into subsets based on feature thresholds, growing a tree structure.
- ▶ Each **split** is chosen to maximize the purity of the resulting subsets (minimize impurity).
- ▶ **Advantages:** Interpretability, handles mixed feature types, no need for feature scaling.
- ▶ **Drawbacks:** Can overfit if not pruned, can be unstable (high variance) for small data.

- ▶ Constructed *recursively*:
 - ▶ Choose a feature + threshold to best split the data (based on impurity).
 - ▶ Split the data into two (or more) subsets.
 - ▶ Repeat the process on each subset until a stopping criterion is met.

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 - ▶ Choose a feature + threshold to best split the data (based on impurity).
 - ▶ Split the data into two (or more) subsets.
 - ▶ Repeat the process on each subset until a stopping criterion is met.
- ▶ Stopping criteria:
 - ▶ Maximum depth reached.
 - ▶ Fewer samples in a node than the minimum required for a split.
 - ▶ Impurity improvement is below a certain threshold.

► **Gini Index (G):**

$$G = \sum_{i=1}^C p_i(1 - p_i) = 1 - \sum_{i=1}^C p_i^2$$

- Measures the chance of misclassification if you pick a label at random.
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► **Information Gain:** The reduction in impurity from a split.